

NAME

msync – synchronize a file with a memory map

LIBRARY

Standard C library (*libc*, *-lc*)

SYNOPSIS

#include <sys/mman.h>

int **msync**(**void** *addr* [*.length*], **size_t** *length*, **int** *flags*);

DESCRIPTION

msync() flushes changes made to the in-core copy of a file that was mapped into memory using **mmap**(2) back to the filesystem. Without use of this call, there is no guarantee that changes are written back before **munmap**(2) is called. To be more precise, the part of the file that corresponds to the memory area starting at *addr* and having length *length* is updated.

The *flags* argument should specify exactly one of **MS_ASYNC** and **MS_SYNC**, and may additionally include the **MS_INVALIDATE** bit. These bits have the following meanings:

MS_ASYNC

Specifies that an update be scheduled, but the call returns immediately.

MS_SYNC

Requests an update and waits for it to complete.

MS_INVALIDATE

Asks to invalidate other mappings of the same file (so that they can be updated with the fresh values just written).

RETURN VALUE

On success, zero is returned. On error, *-1* is returned, and *errno* is set to indicate the error.

ERRORS**EBUSY**

MS_INVALIDATE was specified in *flags*, and a memory lock exists for the specified address range.

EINVAL

addr is not a multiple of **PAGESIZE**; or any bit other than **MS_ASYNC** | **MS_INVALIDATE** | **MS_SYNC** is set in *flags*; or both **MS_SYNC** and **MS_ASYNC** are set in *flags*.

ENOMEM

The indicated memory (or part of it) was not mapped.

VERSIONS

According to POSIX, either **MS_SYNC** or **MS_ASYNC** must be specified in *flags*, and indeed failure to include one of these flags will cause **msync**() to fail on some systems. However, Linux permits a call to **msync**() that specifies neither of these flags, with semantics that are (currently) equivalent to specifying **MS_ASYNC**. (Since Linux 2.6.19, **MS_ASYNC** is in fact a no-op, since the kernel properly tracks dirty pages and flushes them to storage as necessary.) Notwithstanding the Linux behavior, portable, future-proof applications should ensure that they specify either **MS_SYNC** or **MS_ASYNC** in *flags*.

STANDARDS

POSIX.1-2008.

HISTORY

POSIX.1-2001.

This call was introduced in Linux 1.3.21, and then used **EFAULT** instead of **ENOMEM**. In Linux 2.4.19, this was changed to the POSIX value **ENOMEM**.

On POSIX systems on which **msync**() is available, both **_POSIX_MAPPED_FILES** and **_POSIX_SYNCHRONIZED_IO** are defined in <unistd.h> to a value greater than 0. (See also **sysconf**(3).)

SEE ALSO**mmap**(2)

B.O. Gallmeister, POSIX.4, O'Reilly, pp. 128–129 and 389–391.